

Chapter 2, Section 5

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1. Let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{E}$ be a locally free $\mathcal{O}_X$ -module of finite rank. We define the *dual* of $\mathcal{E}$, denoted $\mathcal{E}^\vee$ to be the sheaf $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X)$.

- (a) Show that $(\mathcal{E}^\vee)^\vee \cong \mathcal{E}$.
- (b) For any $\mathcal{O}_X$ -module $\mathcal{F}$, $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{F}) \cong \mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{F}$.
- (c) For any $\mathcal{O}_X$ -modules $\mathcal{F}, \mathcal{G}$, $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E} \otimes \mathcal{F}, \mathcal{G}) \cong \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{G}))$.
- (d) (*Projection Formula*). If $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces, if $\mathcal{F}$ is an $\mathcal{O}_X$ -module, and if $\mathcal{E}$ is a locally free $\mathcal{O}_Y$ -module of finite rank, then there is a natural isomorphism $f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{E}) \cong f_*(\mathcal{F}) \otimes_{\mathcal{O}_Y} \mathcal{E}$.

Lemma 0.1. Let $(X, \mathcal{O})$ be a ringed space, let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}$ -modules, and let $\mathcal{F}$ be locally free of finite rank. For any $x \in X$, $\mathcal{H}om(\mathcal{F}, \mathcal{G})_x \cong \mathcal{H}om_{\mathcal{O}_{x,X}}(\mathcal{F}_x, \mathcal{G}_x)$.

Proof. An element of $\mathcal{H}om(\mathcal{F}, \mathcal{G})_x$ can be represented as a pair $\langle U, f \rangle$, where U is an open neighborhood of x and $f : \mathcal{F}|_U \rightarrow \mathcal{G}|_U$ is a morphism of sheaves on U . Since $(\mathcal{F}|_U)_x \cong \mathcal{F}_x$, $(\mathcal{G}|_U)_x \cong \mathcal{G}_x$, there is a natural map

$$\begin{aligned} \alpha : \mathcal{H}om(\mathcal{F}, \mathcal{G})_x &\rightarrow \mathcal{H}om_{\mathcal{O}_{x,X}}(\mathcal{F}_x, \mathcal{G}_x) \\ \langle U, f \rangle &\mapsto f_x. \end{aligned}$$

We show α is a bijection. If $\mathcal{F}$ has rank n , then $\mathcal{F}_x \cong \mathcal{O}_x^{\oplus n}$ since colimits commute and the image of f_x is a finitely generated $\mathcal{O}_x$ -module. Thus, we can represent f_x as an $n \times n$ -matrix, and there are only a finite number of entries, so we can further assume the entries lie in $\Gamma(W, \mathcal{O})$ for a smaller open neighborhood $W \subseteq U$. It is immediate that α is injective. Also, by this representation any $\varphi \in \mathcal{H}om_{\mathcal{O}_x}(\mathcal{F}_x, \mathcal{G}_x)$ defines a $\mathcal{O}_X(W)$ -module homomorphism $\mathcal{F}(W) = \mathcal{O}_X(W)^{\oplus n} \rightarrow \mathcal{G}(W)$, and by sheafification it ascends to a morphism of sheaves $\mathcal{F}|_W \rightarrow \mathcal{G}|_W$. Hence, α is surjective. $\square$

Lemma 0.2. Let $(X, \mathcal{O}_X)$ be a ringed space, let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$ -modules, and let $x \in X$. Then $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})_x \cong \mathcal{F}_x \otimes_{\mathcal{O}_{x,X}} \mathcal{G}_x$.

Proof. (A.M. Ex. 2.20). $\square$

Lemma 0.3. If $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces, and $\mathcal{F}$ and $\mathcal{G}$ are $\mathcal{O}_Y$ -modules, then $f^*(\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}) \cong f^*(\mathcal{F}) \otimes_{\mathcal{O}_X} f^*(\mathcal{G})$.

Proof. Check locally at each point and use Lemma 2. $\square$

Proof.

- (a) The question is local, so assume $\mathcal{E}$ is free of rank n . Then $\mathcal{E}_x = \mathcal{O}_{x,X}^{\oplus n}$, and by lemma 1, we have

$$(\mathcal{E}^\vee)_x^\vee = \mathcal{H}om_{\mathcal{O}_{x,X}}(\mathcal{H}om_{\mathcal{O}_{x,X}}(\mathcal{O}_{x,X}^{\oplus n}, \mathcal{O}_{x,X}), \mathcal{O}_{x,X}) = (\mathcal{O}_{x,X}^{\oplus n})^\vee \cong \mathcal{O}_{x,X}^{\oplus n} = \mathcal{E}_x,$$

where the last isomorphism is a basic fact of finite free modules. The sheaves $(\mathcal{E}^\vee)^\vee$ and $\mathcal{E}$ are isomorphic at the level of stalks, hence they are isomorphic.

(b) The question is local, so assume $\mathcal{E}$ is free of rank n . For any open set $U \subseteq X$, we have

$$\begin{aligned}\mathcal{H}om_X(\mathcal{E}, \mathcal{F})(U) &= \mathcal{H}om_{\mathcal{O}_U}(\mathcal{E}|_U, \mathcal{F}|_U), \\ (\mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{F})(U) &= \mathcal{E}^\vee(U) \otimes_{\mathcal{O}_U} \mathcal{F}(U) = \mathcal{H}om_{\mathcal{O}_U}(\mathcal{E}|_U, \mathcal{O}_X|_U) \otimes_{\mathcal{O}_U} \mathcal{F}(U),\end{aligned}$$

and since $\mathcal{E}$ is free, we have $\mathcal{E}(U) = \mathcal{O}_X(U)^{\oplus n}$. Let $f_i : \mathcal{E} \rightarrow \mathcal{O}_X$ be the projection morphisms onto the i th component, let $e_i|_U$ be the standard basis of $\mathcal{E}(U)$, and let $f : \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X) \rightarrow \mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{F}$ be the morphism of sheaves defined by

$$\begin{aligned}f(U) : \mathcal{H}om_X(\mathcal{E}, \mathcal{F})(U) &\rightarrow (\mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{F})(U) \\ g &\mapsto \sum_{i=1}^n f_i|_U \otimes g(e_i|_U)\end{aligned}$$

It is isomorphism of sheaves from the fact that $\mathcal{E}$ is free and g is $\mathcal{O}_X$ -linear.

(c) Let $f \in \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E} \otimes \mathcal{F}, \mathcal{G})$, which for every open $U \subseteq X$ defines a homomorphism of abelian groups

$$f(U) : (\mathcal{E} \otimes \mathcal{F})(U) \rightarrow \mathcal{G}(U).$$

We want to show f naturally defines an element of $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{G}))$. Define $\tilde{f} : \mathcal{F} \rightarrow \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{G})$ as the following: for each $s \in \mathcal{F}(U)$, let $\tilde{f}(s) : \mathcal{E}|_U \rightarrow \mathcal{G}|_U$ be the morphism defined by

$$\begin{aligned}\tilde{f}(s) : \mathcal{E}|_U &\rightarrow \mathcal{G}|_U \\ t \in \Gamma(V, \mathcal{E}|_U) &\mapsto f(t \otimes s|_V)\end{aligned}$$

where V is any open subset of U . The fact that this defines an isomorphism of abelian groups follows the corresponding algebraic fact about finite free modules over rings.

(d) The question is local on Y , so assume $\mathcal{E}$ is free of finite rank. Also, if the statement is true for two sheaves of $\mathcal{O}_Y$ -modules, then it true for their direct sum, because left-adjoint functors preserve colimits, so we can assume $\mathcal{E} = \mathcal{O}_Y$. Thus, we have the following composite of canonical isomorphisms

$$f_* \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_Y \cong f_* \mathcal{F} \cong f_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X) \cong f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^* \mathcal{O}_Y).$$

The last isomorphism is because structure sheaves pullback to structure sheaves, i.e.,

$$f^* \mathcal{O}_Y = f^{-1} \mathcal{O}_Y \otimes_{f^{-1} \mathcal{O}_Y} \mathcal{O}_X \cong \mathcal{O}_X.$$

□

7. Let X be a noetherian scheme, and let $\mathcal{F}$ be a coherent sheaf.

- (a) If the stalk $\mathcal{F}_x$ is a free $\mathcal{O}_x$ -module for some point $x \in X$, then there is a neighborhood U of x such that $\mathcal{F}|_U$ is free.
- (b) $\mathcal{F}$ is locally free if and only if its stalks $\mathcal{F}_x$ are free $\mathcal{O}_x$ -modules for all $x \in X$.
- (c) $\mathcal{F}$ is invertible (i.e. locally free of rank 1) if and only if there is a coherent sheaf $\mathcal{G}$ such that $\mathcal{F} \otimes \mathcal{G} \cong \mathcal{O}_X$.

Proof.

- (a) This is a local question, so assume $X = \text{Spec } A$ and $\mathcal{F} = \tilde{M}$ for a noetherian ring A and a finitely generated A -module M . We reduce to the following algebraic problem: if there exists a prime ideal $\mathfrak{p}$ such that $M_{\mathfrak{p}} \cong M \otimes_A A_{\mathfrak{p}}$ is a finite free $A_{\mathfrak{p}}$ -module, then there exists $f \in A - \mathfrak{p}$ such that $M_f \cong M \otimes_A A_f$ is a finite free A_f -module. Let $m_1, \dots, m_r \in M$ such that the image of m_i form a basis in $M_{\mathfrak{p}}$, and let $\phi : A^{\oplus r} \rightarrow M$ be an A -module homomorphism defined by $e_i \mapsto m_i$, where e_i is the standard basis of $A^{\oplus r}$. We have an exact sequence

$$0 \rightarrow \ker \phi \rightarrow A^{\oplus r} \rightarrow M \rightarrow M/\text{im } \phi \rightarrow 0,$$

and localization is an exact functor, so we have an induced exact sequence of $A_{\mathfrak{p}}$ -modules

$$0 \rightarrow (\ker \phi)_{\mathfrak{p}} \rightarrow A_{\mathfrak{p}}^{\oplus r} \rightarrow M_{\mathfrak{p}} \rightarrow (M/\text{im } \phi)_{\mathfrak{p}} \rightarrow 0.$$

Since $A_{\mathfrak{p}}^{\oplus r} \rightarrow M_{\mathfrak{p}}$ is an isomorphism, we have $(\ker \phi)_{\mathfrak{p}}, (M/\text{im } \phi)_{\mathfrak{p}} = 0$. Submodules and quotients of noetherian modules are noetherian; in particular they are finitely generated, so by (A.M. Ex. 2.1) we can find $f \in A - \mathfrak{p}$ such that $(\ker \phi)_f, (M/\text{im } \phi)_f = 0$. Hence, $M_f \cong A_f^{\oplus r}$.

- (b) The if direction follows from (a). The converse direction follows from the following facts: we can realize finite free modules as a colimit, the stalk of a sheaf is defined as a colimit, and colimits commute.
- (c) By the previous parts and lemma 2, we reduce to the following algebraic problem: let A be a noetherian local ring with maximal ideal $\mathfrak{m}$, and let M be a finitely generated A -module. Then $M \cong A$ if and only if there exists a finitely generated A -module N such that $M \otimes_A N \cong A$. One direction is clear. Conversely, suppose M, N are finitely generated A -modules such that $M \otimes_A N \cong A$. Let $k = A/\mathfrak{m}$ be the residue field of A . Tensoring with k gives

$$(M \otimes_A k) \otimes_k (N \otimes_A k) \cong (M \otimes_A N) \otimes_A k \cong A \otimes_A k \cong k,$$

which implies $M \otimes_A k \cong k$. By Nakayama's lemma, M is generated by a single element, which implies $M \cong A/\mathfrak{a}$ for some ideal $\mathfrak{a}$ of A , and similarly $N \cong A/\mathfrak{b}$. We have

$$A \cong M \otimes_A N \cong A/\mathfrak{a} \otimes_A A/\mathfrak{b} \cong A/(\mathfrak{a} + \mathfrak{b}),$$

which implies $\mathfrak{a} + \mathfrak{b} = (0)$. Hence, $M, N \cong A$.

□

8. Again let X be a noetherian scheme, and $\mathcal{F}$ a coherent sheaf on X . We will consider the function

$$\varphi(x) = \dim_{k(x)} \mathcal{F}_x \otimes_{\mathcal{O}_x} k(x),$$

where $k(x) = \mathcal{O}_x/\mathfrak{m}_x$ is the residue field at the point x . Use Nakayama's lemma to prove the following results:

- (a) The function φ is *upper semi-continuous*, i.e. for any $n \in \mathbb{Z}$, the set $\{x \in X \mid \varphi(x) \geq n\}$ is closed.
- (b) If $\mathcal{F}$ is locally free, and X is connected, then φ is a constant function.
- (c) Conversely, if X is reduced, and φ is constant, then $\mathcal{F}$ is locally free.

Lemma 0.4 (Nakayama). *Let A be a local ring with residue field k , and let M be a finitely generated A -module. Then any k -basis of $M/\mathfrak{m}M$ lifts to a minimal set of generators of M .*

Proof.

- (a) We show the set $\{x \in X : \varphi(x) < n\}$ is open. By (5.4), we reduce to the following algebraic problem: let A be a noetherian ring, let M be a finitely generated A -module, and let $\mathfrak{p}$ be a prime ideal of A with residue field $k(\mathfrak{p})$. If $\dim_{k(\mathfrak{p})} M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} k(\mathfrak{p}) < n$ for some $n \in \mathbb{Z}$, there exists a basic open neighborhood $\text{Spec } A_s$ of $\mathfrak{p}$ such that $\dim_{k(\mathfrak{q})} M_{\mathfrak{q}} \otimes_{A_{\mathfrak{q}}} k(\mathfrak{q}) < n$ for all $\mathfrak{q} \in \text{Spec } A_s$, where $s \in A - \mathfrak{p}$. By Nakayama's lemma, there exists $m_1, \dots, m_r \in M$ with $r < n$ such that their image in $M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} k(\mathfrak{p})$ is a $k(\mathfrak{p})$ -basis. Let $f : A^{\oplus r} \rightarrow M$ be the A -module homomorphism defined by $e_i \mapsto m_i$. We have an exact sequence

$$A^{\oplus r} \rightarrow M \rightarrow M/\text{im } f.$$

Localizing gives

$$A_{\mathfrak{p}}^{\oplus r} \rightarrow M_{\mathfrak{p}} \rightarrow (M/\text{im } f)_{\mathfrak{p}},$$

where $(M/\text{im } f)_{\mathfrak{p}} = 0$. Quotients of noetherian modules are noetherian, so by (A.M. Ex. 2.1), we can find $s \in A - \mathfrak{p}$ such that $(M/\text{im } f)_s = 0$. Thus, we have an exact sequence

$$A_s^{\oplus r} \rightarrow M_s \rightarrow 0,$$

which implies $M_{\mathfrak{q}} \otimes_{A_{\mathfrak{q}}} k(\mathfrak{q})$ has rank at most r for any $\mathfrak{q} \in \text{Spec } A_s$.

- (b) Let U_i for $i = 1, \dots, n$ be a finite open cover of X such that $\mathcal{F}|_{U_i}$ is a free $\mathcal{O}_X|_{U_i}$ -module. If X is connected, then for all U_i , there exists U_j with $j \neq i$ such that $U_i \cap U_j$. Clearly φ is constant on each U_i , so let r_i be the rank of $\mathcal{F}|_{U_i}$. Choose any $x \in U_i \cap U_j$, then $r_i = r_j = \varphi(x)$.
- (c) Let $x \in X$, let $\text{Spec } A$ be an open affine neighborhood of x for some noetherian reduced ring A , and let $\mathfrak{p}$ be the prime ideal corresponding to $x \in \text{Spec } A$. Set $r = \varphi(x)$. By (5.4), there exists a finitely generated A -module M such that $\mathcal{F}|_{\text{Spec } A} = \tilde{M}$. Let $m_1, \dots, m_r \in \mathcal{F}_x \cong M_{\mathfrak{p}}$ such that their image in $\mathcal{F}_x \otimes_{\mathcal{O}_x} k(x)$ form a $k(x)$ -basis. By Nakayama's lemma, the m_i 's generate $M_{\mathfrak{p}}$ over $A_{\mathfrak{p}}$, so they generate $M_{\mathfrak{q}}$ for all prime ideals $\mathfrak{q} \subseteq \mathfrak{p}$. Suppose $\sum_{i=1}^r a_i m_i = 0$ for $a_i \in A_{\mathfrak{p}}$. Since the image of the m_i in $M_{\mathfrak{q}} \otimes_{A_{\mathfrak{q}}} k(\mathfrak{q})$ for all $i = 1, \dots, r$ form a $k(\mathfrak{q})$ -basis, the images of a_i in $k(\mathfrak{q})$ must be zero for all i . This implies a_i is contained in all prime ideals of $A_{\mathfrak{p}}$. However, $A_{\mathfrak{p}}$ is reduced, so $a_i = 0$. Hence, m_i are linearly independent.

□

9. Let S be a graded ring, generated by S_1 as an S_0 -algebra, let M be a graded S -module, and let $X = \text{Proj } S$.

- (a) Show that there is a natural homomorphism $\alpha : M \rightarrow \Gamma_*(\tilde{M})$.
- (b) Assume now that $S_0 = A$ is a finitely generated k -algebra for some field k , that S_1 is a finitely generated A -module, and that M is a finitely generated S -module. Show that the map α is an isomorphism in all large enough degrees, i.e. there is a $d_0 \in \mathbb{Z}$ such that for all $d \geq d_0$, $\alpha_d : M_d \rightarrow \Gamma(X, \tilde{M}(d))$ is an isomorphism.
- (c) With the same hypotheses, we define an equivalence relation $\approx$ on graded S -modules by saying $M \approx M'$ if there is an integer d such that $M_{\geq d} \cong M'_{\geq d}$. Here $M_{\geq d} = \bigoplus_{n \geq d} M_n$. We will say that a graded S -module M is *quasi-finitely generated* if it is equivalent to a finitely generated module. Now show that the functors $\sim$ and Γ_* induce an equivalence of categories between the category of quasi-finitely generated graded S -modules modulo the equivalence relation $\approx$, and the category of coherent $\mathcal{O}_X$ -modules.

Proof.

- (a) Write $M = \bigoplus_{d=0}^{\infty} M_d$. If $s \in M_d$, then s determines in a natural way a global section $s \in \Gamma(X, \tilde{M}(d))$, so define $\alpha_d : M_d \rightarrow \Gamma(X, \tilde{M}(d))$ in this way. Define β by extending this map linearly for all d .
- (b) By (I, 7.4), there is a finite filtration

$$0 = M^0 \subseteq M^1 \subseteq \cdots \subseteq M^r = M$$

of M by graded submodules, where for each i , $M^i/M^{i-1} \cong (S/\mathfrak{p}_i)(n_i)$ for some homogenous prime ideal $\mathfrak{p}_i \subseteq S$, and some integer n_i . This filtration gives a filtration of $\tilde{M}$ and short exact sequences

$$0 \rightarrow \tilde{M}^{i-1} \rightarrow \tilde{M}^i \rightarrow \widetilde{M^i/M^{i-1}} \rightarrow 0.$$

Twisting by d and taking global sections, all maps are natural, so we have the following commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & M_d^{i-1} & \longrightarrow & M_d^i & \longrightarrow & (M^i/M^{i-1})_d \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \Gamma(X, \tilde{M}^{i-1}(d)) & \longrightarrow & \Gamma(X, \tilde{M}^i(d)) & \longrightarrow & \Gamma(X, \widetilde{M^i/M^{i-1}}(d)) \end{array}$$

where the vertical arrows are maps defined in (a). By the five lemma, to show that $M_d^i \rightarrow \Gamma(X, \tilde{M}^i(d))$ is surjective for large enough d , it will be sufficient to show that $(S/\mathfrak{p})_d \rightarrow \Gamma(X, \widetilde{S/\mathfrak{p}}(d))$ is surjective for large enough d , for each $\mathfrak{p}$ and n . Thus, we have reduced to the following special case: Let S be a graded integral domain, finitely generated by S_1 as an S_0 -algebra, where $S_0 = A$ is a finitely generated domain over k , and let $X = \text{Proj } S$. Then the map $\alpha : S \rightarrow S' = \bigoplus_{d \geq 0} \Gamma(X, \mathcal{O}_X(d))$ is an isomorphism in all large enough degrees.

Let $x_0, \dots, x_r \in S_1$ be a set of generators of S_1 as an A -module. Following the argument in (5.13), S' is a ring, containing S , and contained in the intersection $\bigcap S_{x_i}$ of the localizations of S at the elements $x_0, \dots, x_r$. By the proof of (5.19), since S'_d is a finitely generated A -module for every d , there exists $d_0 \geq 0$ such that $S_n S'_d \subseteq S_{n+d} \subseteq S'_{n+d}$ for some large enough n and all $d \geq d_0$. It follows by $S_n S'_d = S'_{n+d}$ that $S_{n+d} = S'_{n+d}$ for all $d \geq d_0$.

- (c) The natural homomorphism α viewed as a morphism between equivalence classes of quasi-finitely generated graded S -modules is an isomorphism by (b). Hence, the two categories are equivalent with β from (5.15).

□

12. (a) Let X be a scheme over a scheme Y , and let $\mathcal{L}, \mathcal{M}$ be two very ample invertible sheaves on X . Show that $\mathcal{L} \otimes \mathcal{M}$ is also very ample.
- (b) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms of schemes. Let $\mathcal{L}$ be a very ample invertible sheaf on X relative to Y , and let $\mathcal{M}$ be a very ample invertible sheaf on Y relative to Z . Show that $\mathcal{L} \otimes f^* \mathcal{M}$ is a very ample invertible sheaf on X relative to Z .

Proof.

- (a) Let $g : X \rightarrow \mathbb{P}_Y^r, h : X \rightarrow \mathbb{P}_Y^s$ be immersions such that $\mathcal{L} \cong g^*(\mathcal{O}_{\mathbb{P}_Y^r}(1)), \mathcal{M} \cong h^*(\mathcal{O}_{\mathbb{P}_Y^s}(1))$ respectively, let $\psi : \mathbb{P}_Y^r \times \mathbb{P}_Y^s \rightarrow \mathbb{P}_Y^N$ be a Segre embedding with $N = rs + r + s$, let $\pi_r, \pi_s : \mathbb{P}_Y^r \times \mathbb{P}_Y^s \rightarrow \mathbb{P}_Y^r, \mathbb{P}_Y^s$ be the natural projection morphisms, and let $f : X \rightarrow \mathbb{P}_Y^r \times \mathbb{P}_Y^s$ be the unique morphism such that $g = \pi_r \circ f, h = \pi_s \circ f$. A Segre embedding is an immersion, and compositions of immersions are immersions. In particular, $\psi \circ f$ is an immersion, and we have the following isomorphisms

$$\begin{aligned}
\mathcal{L} \otimes \mathcal{M} &\cong g^*(\mathcal{O}_{\mathbb{P}_Y^r}(1)) \otimes h^*(\mathcal{O}_{\mathbb{P}_Y^s}(1)) \\
&\cong (p_r \circ f)^*(\mathcal{O}_{\mathbb{P}_Y^r}(1)) \otimes (p_s \circ f)^*(\mathcal{O}_{\mathbb{P}_Y^s}(1)) \\
&\cong f^*(p_r^*(\mathcal{O}_{\mathbb{P}_Y^r}(1)) \otimes p_s^*(\mathcal{O}_{\mathbb{P}_Y^s}(1))) \\
&\cong f^*(\mathcal{O}_{\mathbb{P}_Y^r \times \mathbb{P}_Y^s}(1)) \\
&\cong f^*(\psi^*(\mathcal{O}_{\mathbb{P}_Y^N}(1))) \\
&\cong (\psi \circ f)^*(\mathcal{O}_{\mathbb{P}_Y^N}(1)).
\end{aligned}$$

Hence, $\mathcal{L} \otimes \mathcal{M}$ is very ample.

- (b) Let $i : X \rightarrow \mathbb{P}_Y^r$ be an immersion such that $\mathcal{L} \cong i^*\mathcal{O}(1)$, let $j : Y \rightarrow \mathbb{P}_Z^s$ be an immersion such that $\mathcal{M} \cong j^*\mathcal{O}(1)$, and let $\psi : \mathbb{P}_Z^r \times \mathbb{P}_Z^s \rightarrow \mathbb{P}_Z^N$ be a Segre embedding, where $N = rs + r + s$. We have the following commutative diagram

$$\begin{array}{ccccccc}
& & & & & & \mathbb{P}_Z^r \\
& & & & & \nearrow & \\
X & \xrightarrow{i} & \mathbb{P}_Y^r & = & Y \times \mathbb{P}_Z^r & \xrightarrow{j \times 1_{\mathbb{P}_Z^r}} & \mathbb{P}_Z^s \times \mathbb{P}_Z^r = Z \times (\mathbb{P}_Z^r \times \mathbb{P}_Z^s) \xrightarrow{1_Z \times \psi} \mathbb{P}_Z^N = Z \times \mathbb{P}_Z^N \\
& \searrow f & \downarrow & & \downarrow & & \downarrow \\
& & Y & \xrightarrow{j} & \mathbb{P}_Z^s & &
\end{array}$$

where $p : Y \times \mathbb{P}_Z^r \rightarrow Y, q : \mathbb{P}_Z^s \times \mathbb{P}_Z^r \rightarrow \mathbb{P}_Z^s$ are the natural projection maps. A composition of immersions is an immersion, and a product of immersions is an immersion, so $\phi = (1_Z \times \psi) \circ (j \times 1_{\mathbb{P}_Z^r}) \circ i$ is an immersion. Hence, $\mathcal{L} \otimes f^*\mathcal{M} \cong \phi^*\mathcal{O}(1)$. □

13. Let S be a graded ring, generated by S_1 as an S_0 -algebra. For any integer $d > 0$, let $S^{(d)}$ be the graded ring $\bigoplus_{n \geq 0} S_n^{(d)}$ where $S_n^{(d)} = S_{nd}$. Let $X = \text{Proj } S$. Show that $\text{Proj } S^{(d)} \cong X$, and that the sheaf $\mathcal{O}(1)$ on $\text{Proj } S^{(d)}$ corresponds via this isomorphism to $\mathcal{O}_X(d)$.

Proof. The inclusion $i : S^{(d)} \hookrightarrow S$ induces a morphism of schemes $\phi : X \rightarrow X' = \text{Proj } S^{(d)}$ via $\mathfrak{p} \mapsto \mathfrak{p} \cap S^{(d)}$. If $\mathfrak{p} \in X$ so that $\mathfrak{p} \not\supseteq S_+ = \bigoplus_{n > 0} S_n$, then $\mathfrak{p} \not\supseteq S_+^{(d)}$. Otherwise, $s^d \in \mathfrak{p}$ for all $s \in S_+$, which implies $S_+ \supseteq \mathfrak{p}$, a contradiction. Thus, ϕ is well-defined. Since $S^{(d)}$ is generated by $S_1^{(d)}$ as an S_0 -algebra, we can cover X' by open affines $\text{Spec } S_{(f)}^{(d)}$, where $f \in S_1^{(d)} = S_d$ and $S_{(f)}^{(d)}$ consists of all degree zero elements of the form s/f^n for some $s \in S^{(d)}$. Also, $\phi^{-1}(\text{Spec } S_{(f)}^{(d)}) = \text{Spec } S_{(f)}$, and clearly $S_{(f)} \cong S_{(f)}^{(d)}$ as rings. Hence, ϕ is an isomorphism of schemes. Lastly, we also have an isomorphism of twisted sheaves $\mathcal{O}_{X'}(1) \cong (i_*\mathcal{O}_X)(1) = i_*(\mathcal{O}_X(d))$. □

16. Let $(X, \mathcal{O}_X)$ be a ringed spaces, and let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. We define the *tensor algebra*, *symmetric algebra*, and *exterior algebra* of $\mathcal{F}$ by taking the sheaves associated to the presheaf, which to each open set U assigns the corresponding tensor operation applied to $\mathcal{F}(U)$ as an $\mathcal{O}_X(U)$ -module. The results are $\mathcal{O}_X$ -algebras, and their components in each degree are $\mathcal{O}_X$ -modules.

- (a) Suppose that $\mathcal{F}$ is locally free of rank n . Then $T^r(\mathcal{F})$, $S^r(\mathcal{F})$, and $\bigwedge^r(\mathcal{F})$ are also locally free, of ranks n^r , $\binom{n+r-1}{r}$, and $\binom{n}{r}$ respectively.
- (b) Again let $\mathcal{F}$ be locally free of rank n . Then the multiplication map $\bigwedge^r \mathcal{F} \otimes \bigwedge^{n-r} \mathcal{F} \rightarrow \bigwedge^n \mathcal{F}$ is a perfect pairing for any r , i.e. it induces an isomorphism of $\bigwedge^r \mathcal{F}$ with $(\bigwedge^{n-r} \mathcal{F})^\vee \otimes \bigwedge^n \mathcal{F}$. As a special case, note if $\mathcal{F}$ has rank 2, then $\mathcal{F} \cong \mathcal{F}^\vee \otimes \bigwedge^2 \mathcal{F}$.

- (c) Let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ be an exact sequence of locally free sheaves. Then for any r there is a finite filtration of $S^r(\mathcal{F})$,

$$S^r(\mathcal{F}) = F^0 \supseteq F^1 \supseteq \dots \supseteq F^r \supseteq F^{r+1} = 0$$

with quotients

$$F^p / F^{p+1} \cong S^p(\mathcal{F}') \otimes S^{r-p}(\mathcal{F}'')$$

for each p .

- (d) Same statement as (c), with exterior powers instead of symmetric powers. In particular, if $\mathcal{F}', \mathcal{F}, \mathcal{F}''$ have ranks n', n, n'' respectively, there is an isomorphism $\bigwedge^n \mathcal{F} \cong \bigwedge^{n'} \mathcal{F}' \otimes \bigwedge^{n''} \mathcal{F}''$.
- (e) Let $f : X \rightarrow Y$ be a morphism of ringed spaces, and let $\mathcal{F}$ be an $\mathcal{O}_Y$ -module. Then f^* commutes with all the tensor operations on $\mathcal{F}$, i.e. $f^*(S^n(\mathcal{F})) = S^n(f^*\mathcal{F})$.

Proof.

- (a) The question is local, so the statements follow from the case for rings. For example, let U be an open set in X such that $\mathcal{F}|_U \cong \mathcal{O}_U^{\oplus n}$. Then $T^r(\mathcal{F})|_U \cong T^r(\mathcal{O}_U^{\oplus n}) \cong \mathcal{O}_U^{\oplus n^r}$. We remark that locally, the symmetric algebra $S(\mathcal{F})$ is isomorphic to the polynomial ring $\mathcal{O}[T_1, \dots, T_n]$, and $S^r(\mathcal{F})$ correspond to the homogenous elements of degree r in $\mathcal{O}[T_1, \dots, T_n]$. Thus, $S^r(\mathcal{F})$ has rank $\binom{n+r-1}{r}$. Let $x_1, \dots, x_n \in \Gamma(U, \mathcal{O}_X)$ be a $\mathcal{O}_U$ -basis for $\mathcal{F}|_U$. Then $\bigwedge^r(\mathcal{F})$ is spanned by $x_{i_1} \wedge \dots \wedge x_{i_r}$, where $1 \leq i_1 < \dots < i_r \leq n$. Hence, $\bigwedge^r(\mathcal{F})$ has rank $\binom{n}{r}$.

□